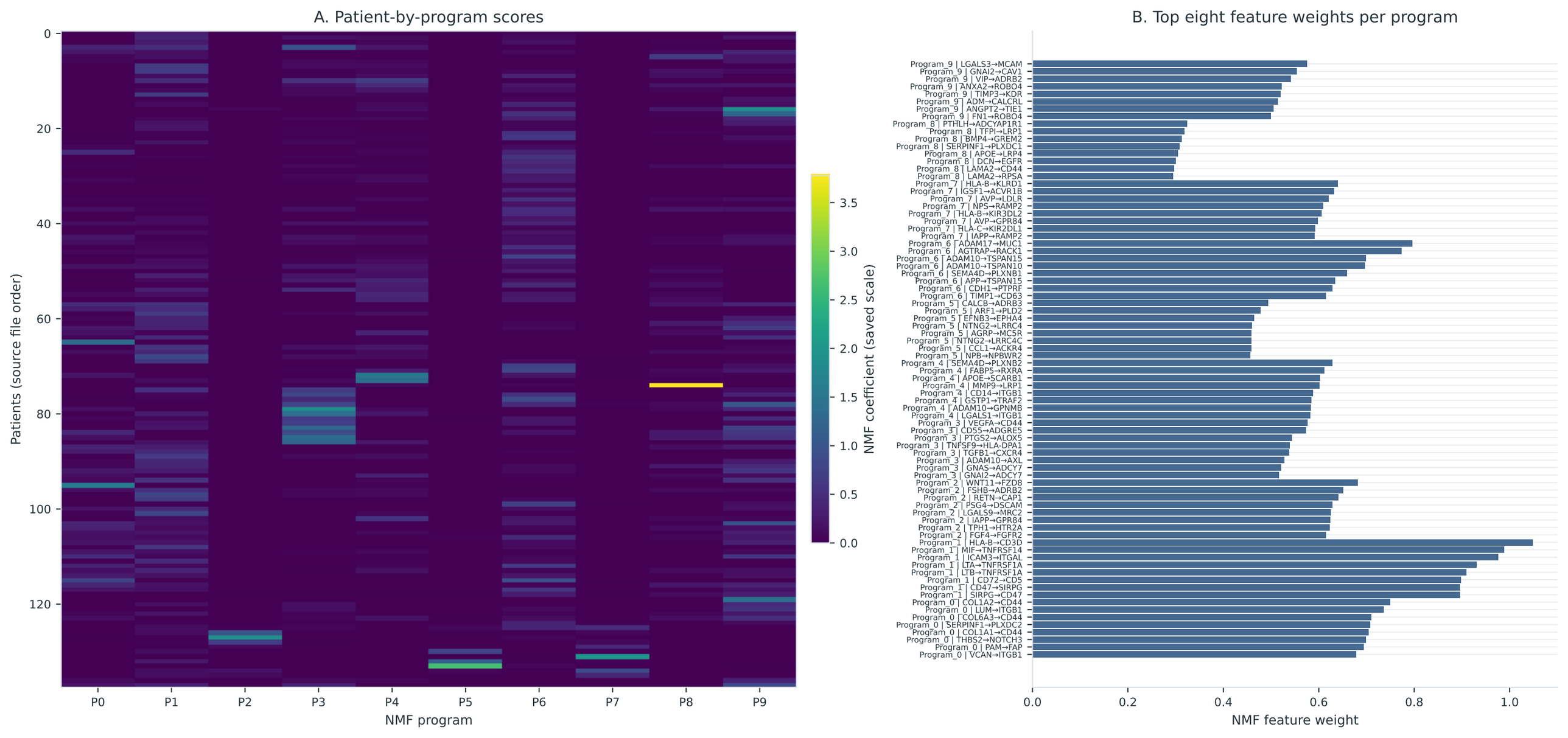
A. Patient-by-program scores
Patients (source file order)
0
20
40
60
80
100
120
P0
P1
P2
P3
P4
P5
P6
P7
P8
P9
NMF program
NMF coefficient (saved scale)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
B. Top eight feature weights per program
Program_9 | LGALS3→MCAM
Program_9 | GNAI2→CAV1
Program_9 | VIP→ADRB2
Program_9 | ANXA2→ROBO4
Program_9 | TIMP3→KDR
Program_9 | ADM→CALCRL
Program_9 | ANGPT2→TIE1
Program_9 | FN1→ROBO4
Program_8 | PTHLH→ADCYAP1R1
Program_8 | TFPI→LRP1
Program_8 | BMP4→GREM2
Program_8 | SERPINF1→PLXDC1
Program_8 | APOE→LRP4
Program_8 | DCN→EGFR
Program_8 | LAMA2→CD44
Program_8 | LAMA2→RPSA
Program_7 | HLA-B→KLRD1
Program_7 | IGSF1→ACVR1B
Program_7 | AVP→LDLR
Program_7 | NPS→RAMP2
Program_7 | HLA-B→KIR3DL2
Program_7 | AVP→GPR84
Program_7 | HLA-C→KIR2DL1
Program_7 | IAPP→RAMP2
Program_6 | ADAM17→MUC1
Program_6 | AGTRAP→RACK1
Program_6 | ADAM10→TSPAN15
Program_6 | ADAM10→TSPAN10
Program_6 | SEMA4D→PLXNB1
Program_6 | APP→TSPAN15
Program_6 | CDH1→PTPRF
Program_6 | TIMP1→CD63
Program_5 | CALCB→ADRB3
Program_5 | ARF1→PLD2
Program_5 | EFNB3→EPHA4
Program_5 | NTNG2→LRRC4
Program_5 | AGRP→MC5R
Program_5 | NTNG2→LRRC4C
Program_5 | CCL1→ACKR4
Program_5 | NPB→NPBWR2
Program_4 | SEMA4D→PLXNB2
Program_4 | FABP5→RXRA
Program_4 | APOE→SCARB1
Program_4 | MMP9→LRP1
Program_4 | CD14→ITGB1
Program_4 | GSTP1→TRAF2
Program_4 | ADAM10→GPNMB
Program_4 | LGALS1→ITGB1
Program_3 | VEGFA→CD44
Program_3 | CD55→ADGRE5
Program_3 | PTGS2→ALOX5
Program_3 | TNFSF9→HLA-DPA1
Program_3 | TGFB1→CXCR4
Program_3 | ADAM10→AXL
Program_3 | GNAS→ADCY7
Program_3 | GNAI2→ADCY7
Program_2 | WNT11→FZD8
Program_2 | FSHB→ADRB2
Program_2 | RETN→CAP1
Program_2 | PSG4→DSCAM
Program_2 | LGALS9→MRC2
Program_2 | IAPP→GPR84
Program_2 | TPH1→HTR2A
Program_2 | FGF4→FGFR2
Program_1 | HLA-B→CD3D
Program_1 | MIF→TNFRSF14
Program_1 | ICAM3→ITGAL
Program_1 | LTA→TNFRSF1A
Program_1 | LTB→TNFRSF1A
Program_1 | CD72→CD5
Program_1 | CD47→SIRPG
Program_1 | SIRPG→CD47
Program_0 | COL1A2→CD44
Program_0 | LUM→ITGB1
Program_0 | COL6A3→CD44
Program_0 | SERPINF1→PLXDC2
Program_0 | COL1A1→CD44
Program_0 | THBS2→NOTCH3
Program_0 | PAM→FAP
Program_0 | VCAN→ITGB1
0.0
0.2
0.4
0.6
0.8
1.0
NMF feature weight